

ALEX LU

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Senior Researcher

Microsoft Research New England

RESEARCH INTERESTS

My research interests center around AI for scientific discovery, with a focus on biology:

- I precisely characterize core limitations of current AI for biology efforts.
- I develop biologically-grounded methods that address these limitations to unlock new capabilities.
- I translate insights to widespread impact in AI for science by collaborating with domain experts.

Key recent publications for each topic area are listed under Selected Publications and Pre-Prints. My personal research website can be found at alexluresearch.com.

SELECTED PUBLICATIONS AND PRE-PRINTS

Characterizing limitations of AI for biology

- Zero-shot evaluation reveals limitations of single-cell foundation models.
Kedzierska KZ, Crawford L, Amini AP, **Lu AX**. Genome Biology. 2025 Apr 18.
- Feature Reuse and Scaling: Understanding Transfer Learning with Protein Language Models.
Li FZ, Amini AP, Yue Y, Yang KK, **Lu AX**. ICML 2024.2024 Jul 8.

Biologically-grounded AI methods

- Evolutionary conditioning enables guided generation of functionally diverse enhancers.
Duncan AG, Consens ME, Crawford L, Mitchell JA, Moses AM, Yang KK, **Lu AX**. Preprint. 2026 Apr 15.
- Vermeer: Autoregressive generative modeling of microscopy predicts protein localization.
Kambhampati S, Zimmermann E, Hayir EB, Yang KK, Chen Fei, **Lu AX**. Submitted. 2026 May.
- Protein structure generation via folding diffusion. Wu KE, Yang KK, Van den Berg R, Alamdari S, Zou JY, **Lu AX**, Amini AP. Nature Communications. 2024 Feb 5.

Collaboration enables AI for science

- Underrepresentation of children in public medical imaging datasets.
Hua SB, Heller N, He P, Towbin AJ, Chen IY, **Lu AX**, Erdman L. Nature Health. 2026 Apr 24.
- Systemic biases in sign language AI research: A deaf-led call to reevaluate research agendas.
Desai A, De Meulder M, Hochgesang JA, Kocab A, **Lu AX**. LREC-COLING 2024 11th Workshop on the Representation and Processing of Sign Languages 2024 May.
- Enhancing AI and Dynamical Subseasonal Forecasts with Probabilistic Bias Correction.
Guan H, Mouatadid S, Orenstein P, Cohen J, Dong H, Ni Z, Berman J, Flaspohler G, **Lu AX**, Schloer J, Talib J, Weyn JA, Mackey L. Preprint. 2026 Apr 17.

Peer Reviewed Publications

- Predicting evolutionary rate as a pretraining task improves genome language model representations. Consens ME, Yang KK, Hall J, Conard AM, Wang B, Crawford L, Moses A, **Lu AX**. Accepted at ICML 2026.
- FLIP2: Expanding Protein Fitness Landscape Benchmarks for Real-World Machine Learning Applications. Didi K, Alamdari S, **Lu AX**, Wittmann B, Johnston KE, Amini AP, Madani A, Czeneszew M, Dallago C, Yang KK. Accepted at ICML 2026.
- A functional map of the human intrinsically disordered proteome. Pritisanac I, Alderson TR, Kolarić D, Zarin T, Xie S, **Lu AX**, Alam A, Maqsood A, Youn JY, Forman-Kay JD, Moses AM. In press at PNAS.
- Underrepresentation of children in public medical imaging datasets. Hua SB, Heller N, He P, Towbin AJ, Chen IY, **Lu AX**, Erdman L. Nature Health. 2026 Apr 24.
- scGeneScope: A treatment-matched single cell imaging and transcriptomics dataset and benchmark for treatment response modeling. Dapello J, Nassar M, Eksi R, Wang B, Gagnon-Marchand J, Gao K, Thrush K, Riehs N, Peterson A, Tolpadi A, Rajagopal A, Miller H, Conard A, Alvarez-Melis D, Stark R, Bianco S, Levine M, Amini A, **Lu AX**, Fusi N, Pandya R, Pedoia V, El-Samad H. NeurIPS 2025. 2026 Apr 23.
- Exploring Collaboration to Center the Deaf Community in Sign Language AI. Kamikubo R, Glasser A, **Lu AX**, Daumé III H, Kacorri H, Bragg D. In Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility 2025 Oct 26.
- Investigating Dictionary Expansion for Video-based Sign Language Dictionaries. Desai A, Massiceti D, Ladner R, Daumé III H, Bragg D, **Lu AX**. In Findings of the Association for Computational Linguistics: EMNLP 2025. 2025 Nov.
- Exploring Reduced Feature Sets for American Sign Language Dictionaries. Kosa B, Desai A, **Lu AX**, Ladner RE, Bragg D. In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems 2025 Apr 26.
- Representation Learning Methods for Single-Cell Microscopy are Confounded by Background Cells. Gupta A, Moses A, **Lu AX**. MLCB 2025. 2025 June.
- Zero-shot evaluation reveals limitations of single-cell foundation models. Kedzierska KZ, Crawford L, Amini AP, **Lu AX**. Genome Biology. 2025 Apr 18.
- ASL STEM Wiki: Dataset and Benchmark for Interpreting STEM Articles. Yin K, Singh C, Minakov FO, Milan V, Daumé III H, Zhang C, **Lu AX**, Bragg D. EMNLP 2024. 2024 Nov 8.
- Feature Reuse and Scaling: Understanding Transfer Learning with Protein Language Models. Li FZ, Amini AP, Yue Y, Yang KK, **Lu AX**. ICML 2024. 2024 Jul 8.
- Systemic biases in sign language AI research: A deaf-led call to reevaluate research agendas. Desai A, De Meulder M, Hochgesang JA, Kocab A, **Lu AX**. LREC-COLING 2024 11th Workshop on the Representation and Processing of Sign Languages 2024 May.
- Convolutions are competitive with transformers for protein sequence pretraining. Yang KK, Fusi N, **Lu AX**. Cell Systems. Feb 2024.
- ASL Citizen: A Community-Sourced Dataset for Advancing Isolated Sign Language Recognition. Desai A, Berger L, Minakov FO, Milan V, Singh C, Pumphrey K, Ladner RE, Daumé III H, **Lu AX**, Caselli N, Bragg D. NeurIPS 2023. 2023 Dec.

- Domain adaptation using optimal transport for invariant learning using histopathology datasets. Falahkheirkhah K, **Lu AX**, Alvarez-Melis D, Huynh G. MIDL 2023. 2023 Jul.
- Supervised contrastive learning for improved view labeling in pediatric renal ultrasound videos. Hua SB, Chen I, **Lu AX**, Erdman L. ISBI 2023. 2023 April 18.
- Incorporating knowledge of plates in batch normalization improves generalization of deep learning for microscopy images. Lin A, **Lu AX**. MLCB 2022. 2022 Dec 19.
- Discovering molecular features of intrinsically disordered regions by using evolution for contrastive learning. **Lu AX**, Lu AXP, Pritisanac I, Zarin T, Forman-Kay JD, Moses AM. PLOS Computational Biology. 2022 Jun 29.
- Learning from physics-based features improves protein property prediction. Wang A, Amini AP, **Lu AX**, Yang KK. MLSB 2022.
- CytoImageNet: A large-scale pretraining dataset for bioimage transfer learning. Hua SB, **Lu AX**, Moses AM. LMRL workshop at NeurIPS 2021.
- Improved Conditional Flow Models for Molecule to Image Synthesis. Yang K, Goldman S, Jin W, **Lu A**, Barzilay R, Jaakkola T, Uhler C. CVPR 2021.
- Evolution Is All You Need: Phylogenetic Augmentation for Contrastive Learning. Lu AXP, **Lu AX**, Moses, AM. MLCB 2020.
- The Cells Out of Sample (COOS) dataset and benchmarks for measuring out-of-sample generalization of image classifiers. **Lu AX**, Lu AXP, Schormann W, Andrews DW, Moses AM. NeurIPS 2019. 2019 December.
- Learning unsupervised feature representations for single cell microscopy images with paired cell inpainting. **Lu AX**, Kraus OZ, Cooper S, Moses AM. PLOS Computational Biology. 2019 Sept 3.
- YeastSpotter: accurate and parameter-free web segmentation for microscopy images of yeast cells. Bioinformatics. **Lu AX**, Zarin T, Hsu IS, Moses AM. 2019 May 16.
- Integrating images from multiple microscopy screens reveals diverse patterns of change in the subcellular localization of proteins. **Lu AX**, Chong YT, Hsu IS, Strome B, Handfield LF, Kraus O, Andrews BJ, Moses AM. Elife. 2018 Apr 5.
- Extracting and integrating protein localization changes from multiple image screens of yeast cells. **Lu AX**, Handfield LF, Moses A. Bio-Protocol. 2018 Apr 8.
- Influence of repetitive mechanical loading on MMP2 activity in tendon fibroblasts. Huisman E, **Lu AX**, Jamil S, Mousavizadeh R, McCormack R, Roberts C, Scott A. Journal of Orthopaedic Research. 2016 Nov.
- An unsupervised kNN method to systematically detect changes in protein localization in high-throughput microscopy images. **Lu AX**, Moses AM. PloS One. 2016 Jul 21.
- Angiopoietin-like 4 promotes angiogenesis in the tendon and is increased in cyclically loaded tendon fibroblasts. The Journal of physiology. Mousavizadeh R, Scott A, **Lu AX**, Ardekani GS, Behzad H, Lundgreen K, Ghaffari M, McCormack RG, Duronio V. 2016 Jun 1.
- Accumulation of oxidized LDL in the tendon tissues of C57BL/6 or apolipoprotein E knock-out mice that consume a high fat diet: potential impact on tendon health. Grewal N, Thornton GM, Behzad H, Sharma A, **Lu AX**, Zhang P, Reid WD, Granville DJ, Scott A. PLoS One. 2014 Dec 12.

- Enhanced collagen type I synthesis by human tenocytes subjected to periodic in vitro mechanical stimulation. Huisman E, **Lu AX**, McCormack RG, Scott A. BMC musculoskeletal disorders. 2014 Dec.
- Podocalyxin regulates murine lung vascular permeability by altering endothelial cell adhesion. Debruin EJ, Hughes MR, Sina C, **Lu AX**, Cait J, Jian Z, Lopez M, Lo B, Abraham T, McNagny KM. PloS one. 2014 Oct 10.
- Mast cells exert pro-inflammatory effects of relevance to the pathophysiology of tendinopathy. Behzad H, Sharma A, Mousavizadeh R, **Lu AX**, Scott A. Arthritis research & therapy. 2013 Dec.

Pre-prints and submissions

- Vermeer: Autoregressive generative modeling of microscopy predicts protein localization. Kambhampati S, Zimmermann E, Hayir EB, Yang KK, Chen Fei, **Lu AX**. Submitted. 2026 May.
- MorphoHELM: A Comprehensive Benchmark for Evaluating Representations for Microscopy-Based Morphology Assays. Hayir EB, Crawford L, **Lu AX**. Submitted.
- Phonological Perception of Sign Language Models. Yin K, Carter J, **Lu AX**, Kocab AM. Submitted.
- Evolutionary conditioning enables guided generation of functionally diverse enhancers. Duncan AG, Consens ME, Crawford L, Mitchell JA, Moses AM, Yang KK, **Lu AX**. Preprint. 2026 Apr 15.
- Multimodal Alignment Improves Generalizability of Genomic Biomarker Prediction in Computational Pathology. Redekop E, Zimmermann E, Amini AP, **Lu AX**, Tenenholtz N, Hall JB, Crawford L, Severson KA. Preprint. 2026 Feb 27.
- Enhancing AI and Dynamical Subseasonal Forecasts with Probabilistic Bias Correction. Guan H, Mouatadid S, Orenstein P, Cohen J, Dong H, Ni Z, Berman J, Flaspohler G, **Lu AX**, Schloer J, Talib J, Weyn JA, Mackey L. Preprint. 2026 Apr 17.
- Integrating chemical structures as treatments improves representations of microscopy images for morphological profiling. Yu Y, Hayir EB, Tenenholtz N, Mackey L, Wei Y, Alvarez-Melis D, Amini AP, **Lu AX**. Preprint. 2025 Apr 13.
- Protein generation with evolutionary diffusion: sequence is all you need. Alamdari S, Thakkar N, van den Berg R, **Lu AX**, Fusi N, Amini AP, Yang KK. Preprint. 2023 Sep.

Book Chapters and Other Publications

- Transcription Factors: Disorderly Conduct. Weinstein E, Moses AM, **Lu AX**. GEN Biotechnology. 2026 Apr 7.
- Localization velocity: visualizing changes in protein localization detected in high-throughput microscopy. **Lu AX**, Moses AM. Methods in Molecular Biology - Imaging Cell Signaling (Springer Nature). May 2024.

Invited Talks

- **Beyond Translation:
Why linguistic study of sign language empowers deaf communities**
Keynote at CVPR MLSB Workshop
CVPR, Denver CO USA, June 3 2026
- **Advancing generative genomics models with evolution**
AI in Genomics Symposium
University of Chicago, Chicago IL USA, May 8 2026
- **Advancing generative genomics models with evolution**
Big Data Institute
Oxford University, Oxford UK, April 14 2026
- **Advancing generative genomics models with evolution**
Microsoft Research Cambridge, Cambridge UK, April 13 2026
- **AI to Accelerate Scientific Discovery**
Cognitive Science Department
John Hopkins University, Baltimore, MD, April 2nd 2025
- **Towards Meaningful Pretrained Models for Biology**
Models, Algorithms and Inference, Broad Institute
Broad Institute, Cambridge, MA, October 24th, 2023
- **Understanding big biological data with self-supervised machine learning**
EWSC Initiative, Broad Institute
Broad Institute, Cambridge, MA, March 17th, 2022
- **Understanding big biological data with self-supervised machine learning**
University of Rochester
Virtual Seminar, March 10th, 2022
- **Understanding biomedical images with self-supervised machine learning**
Spring Discovery
Virtual Seminar, October 13th, 2021
- **Learning biology through puzzle-solving: unbiased automatic understanding of microscopy images through self-supervised learning**
Microscopy & Microanalysis 2020
Virtual Conference, August 4th, 2020
- **Learning unsupervised feature representations for single cell microscopy images with paired cell inpainting**
Cell Painting Group, Broad Institute
Virtual Seminar, April 6th, 2020
- **Exploring big image data with self-supervised deep learning**
OICR Seminar Series
Toronto, Canada, November 11th, 2019
- **Learning unsupervised feature representations for single cell microscopy images with paired cell inpainting**
TorBUG Seminar Series
Toronto, Canada, November 28th, 2018

- **Discovering biological insights from large-scale microscopy datasets with unsupervised deep learning**
Sunnybrook Research Institute
Toronto, Canada, November 26th, 2018

Contributed Oral Presentations

- **Systemic Biases in Sign Language AI Research**
LREC-COLING 2024
Turn, Italy, May 2024
- **Discovering interpretable features of the intrinsically disordered dark proteome with contrastive learning**
ISMB 2022
Madison, Wisconsin, July 2022
- **Learning unsupervised feature representations for single cell microscopy images with paired cell inpainting**
MLCB 2019
Vancouver, Canada, December 14th, 2019
- **Learning unsupervised feature representations for single cell microscopy images with paired cell inpainting**
Biological Data Science 2018
Cold Spring Harbor Laboratory, Long Island, New York, November 10th, 2018
- **An unsupervised kNN method to systematically detect changes in protein localization in high-throughput microscopy images**
GLBIO/CCBC 2016
Toronto, Canada, May 16th, 2016

Poster Presentations

- **Transfer Learning vs. Batch Effects: what can we expect from neural networks in computational biology?**
MLCB 2019
Vancouver, Canada, December 14th, 2019
- **The Cells Out of Sample (COOS) dataset and benchmarks for measuring out-of-sample generalization of image classifiers.**
NeurIPS 2019
Vancouver, Canada, December 11th, 2019
- **Learning representations of biology with small and homogeneous training datasets**
NeurIPS 2019 (Learning Meaningful Representations of Life Workshop)
Vancouver, Canada December 13th, 2019
- **Paired cell inpainting: self-supervised multiple-instance learning for bioimage analysis**
ICML 2019 (Self-Supervised Learning Workshop)
Long Beach, California, June 16th, 2019

EDUCATION

PhD (Computer Science)

2017 - 2021

University of Toronto

Thesis: Unsupervised machine learning for hypothesis discovery and representation learning in biological image and sequence analysis.

MSc (Computer Science)

2016 - 2017

University of Toronto

Thesis: An unsupervised kNN method to systematically detect changes in protein localization in high-throughput microscopy images.

BSc (Honors Specialization in Bioinformatics)

2010 - 2015

University of Western Ontario

Thesis: Observations in extracting and parsing subcellular relations using natural language processing systems in biomedical publications.

ACADEMIC WORK EXPERIENCE

Senior Researcher

November 2021 - Present

Microsoft Research New England

Intern

Jan 2019 - May 2019

Phenomic AI, Toronto

Teaching Assistant

Sept 2015 - May 2016

Department of Computer Science, University of Toronto **Bioinformatics Undergraduate Research**

Assistant

May 2015 - August 2015

Robert HN Ho Research Center, University of British Columbia

TEACHING AND SERVICE

Interns

- 2026: Taykhoom Dalal, Arjun Naik
- 2025: Sandeep Kambhampati, Hannah Guan, Ekaterina Redekop
- 2024: Suyue Lyu, Micaela Consens, Chris Lin, Darren Xu
- 2023: Kasia Kedzierska, Yemin Yu, Arushi Gupta, Kayo Yin, Giovanni Palla
- 2022: Francesca Zhoufan-Li, Kevin Wu, Amy Wang, Nitya Thakkar, Alexander Lin, Wenjun Wu, Aashaka Desai, Rie Kamikubo, David Martin, Kianoush Falahkheirkhah

Co-supervised Students at Academic Institutions

Note: I maintain a supervisory relationship with some interns past their internship. I list those who go on to work on new, additional projects beyond their internship work.

- Jianpu Gao (Undergraduate, UC Berkeley, 2025-Present)
- Vatsala Nema (PhD, John Hopkins University, 2025-Present)
- Andrew Duncan (PhD, University of Toronto, 2024-Present)
- Micaela Consens (PhD, University of Toronto, 2024-Present)
- Kayo Yin (PhD, UC Berkeley, 2023-2026)
- Aashaka Desai (PhD, University of Washington, 2023-2025)

- Stanley Hua (BSc, University of Toronto, 2020-2022; Research Assistant, Cincinnati's Children Hospital 2022-2024)
- Riad Dajani (MSc, John Hopkins University, 2023-2025)

Committees

- **Lead Position Paper Track Chair** *Jan 2026 - Present*
NeurIPS
Lead PC for Position Paper Track at NeurIPS 2026
- **Communications Chair** *Feb 2023 - Dec 2025*
NeurIPS
Communications chair for NeurIPS
- **Thesis Dissertation Committee** *Jan 2024 - Present*
University of Washington PhD thesis dissertation member for Aashaka Desai.
- **Microsoft PhD Fellowship Committee** *June 2022 - October 2022*
Microsoft Research Jointly led Health and Life Sciences committee for awarding PhD fellowships - coordinated reviews for over 100 applications.
- **TorBUG Committee Member** *June 2019 - June 2021*
TorBUG, University of Toronto
Committee member for organizing monthly bioinformatics seminar series.

Teaching

- **Collaborator on Bioinformatics Training Platform**
January 2024 to Present
Collaborator on Bioinformatics.ca program intended to aggregate, scale, and link training communities across Canada.
- **Genome Biology and Bioinformatics Program: Deep Learning for Biology Seminars**
November 2019 to January 2020
Designed and taught three-part talk series for the Genome Biology and Bioinformatics program at the University of Toronto.
- **Introduction to Deep Learning for Biologists**
September 17th and 24th, 2018
Designed and taught introductory workshops on building deep neural networks to graduate students in the Department of Cells and Systems Biology at the University of Toronto.

Service

- **Advisor and Keynote Speaker for AIBIDS Program at Gallaudet/University of Pittsburgh**
Dec 2022 to June 2023 - advised biomedical informatics program for underrepresented students.
- **Speaker for ISMB/ECCB - Advancing Accessibility in Bioinformatics Training**
July 2023 - speaker on how to advance diversity and inclusion in relation to disability in bioinformatics.
- **Speaker for Accessibility Services at University of Toronto**
March 29 2023 - advised students on navigating disability and work in panel discussion.
- **Dean's Advisory Council for Arts and Science Faculty at University of Toronto**
April 2022 to 2024 - advised then-Dean Melanie Woodin on academic programming.

- **Peer Review**

Peer reviewer for NeurIPS 2021-2025, Nature Communications, Nature Cell Biology, Nature Methods, Genome Biology, PLOS Computational Biology, MLCB 2020, Bioinformatics, Cell Systems, CIBB 2019, IEEE TM, and BMC Supplements. Area chair for LMRL 2022.

- **Community Engagement**

Speaker for Skype a Scientist for high school classes in Providence, RI and Englewood, NJ.

- **Conference Organizing**

Scientific committee for DAC2021.

Open-Source Software

- **YeastSpotter**

December 2018 to Present

Developer and maintainer of open-source segmentation web application YeastSpotter (<http://yeastspotter.csb.utoronto.ca>)

HONORS AND AWARDS

NSERC MSFSS

Award to fund visiting research position; \$6,000 for three months

Dec 2019

Ray Reiter Graduate Award in Computer Science

Departmental award with value of \$1,000

Dec 2019

MLCB Travel Award

Conference award at MLCB, awarded on basis of merit to five outstanding papers

Dec 2019

NSERC CGS-D

National research fellowship: awarded \$35,000 per year over three years

Ranked 1st of 192 applicants in cellular and molecular biology

May 2017

Best Paper: GLBIO/CCBC 2017

Regional conference award for best full paper

May 2016

NSERC CGS-M

National research fellowship: awarded \$13,000. Declined due to institution choice.

(Declined)

NSERC URSA

National undergraduate research fellowship: awarded \$4,500 for summer research

May 2015

Western Gold Medal for Bioinformatics

Institutional: Awarded to top graduating student from program

May 2015

NSERC URSA

National undergraduate research fellowship: awarded \$4,500 for summer research

May 2013